

Femtosecond-laser-writing in various glasses

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Abstract

The localized modification of the refractive index in different glasses using a femtosecond laser direct-writing technique at 800 nm is described. The induced refractive-index changes were positive and negative depending on glass composition and laser-writing conditions. The optical properties of the generated structures were studied with different methods. Quantitative measurements were possible with a special interference microscope with shearing technique. Large differences were detected depending on glass composition and structure. Selected glasses were fused silica, multi-component borosilicate, borate and fluoride phosphate glasses. The induced refractive change in fluoride phosphate glasses increased with increasing phosphate content up to $\sim 5 \times 10^{-3}$. A redistribution of fluoride and phosphate groups in the local irradiated volume is assumed in correlation with the crystallization behavior of these glasses.

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1. Introduction

The research in glass modification by use of ultra-short laser pulses is driven by scientific interest and by potential applications for fabricating three-dimensional photonic devices for integrated optics. Today well-known technologies, like ion-exchange, diffusion into a transparent substrate or lithographic methods, are generally limited to the fabrication of planar structures. Recently, a novel direct technique based on the use of ultra-short laser pulses has been demonstrated [1–6]. When these intensive laser pulses are focused into the bulk of a transparent material, non-linear absorption in the focal volume takes place. Femtosecond-laser induced changes of the refractive index are due to multi-photon processes by non-linear absorption. At sufficient high laser pulse intensities a local redistribution of the

glass structure in the focal volume takes place which leads consequently to a mainly permanent modification of the microstructure and refractive index. Depending on the precise laser conditions – wavelength, pulse energy, pulse duration, repetition rate, exposure time – the induced changes can range from merely a permanent change in refractive index to macroscopic crack formation. Most studies have focused on the changes in fused silica because of its wide use as an optical material with a well-known structure [3]. It was demonstrated that by changing only the fs-writing speed, waveguides with a controllable mode number can be generated [6]. There are contradictory reports of the mechanisms and of the magnitude of the induced refractive change (Δn). Nearly nothing is known about the dependence on the glass composition and structure. In this paper we describe our first results of some quantitative experiments with selected different glasses which were fs-laser-written under the same conditions (Table 1). The local structure of these glasses, their crystallization behavior, color center formation and other properties were extensively studied before [7–12]. The aim is the detection of a

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Table 1
Glass compositions and properties

	Composition (mol%)						
	SiO ₂ (100SiO ₂)	UBK7 (74SiO ₂ 10B ₂ O ₃ 15Na ₂ O/K ₂ O 1BaO)	UV-DURAN (82SiO ₂ 12B ₂ O ₃ 5Na ₂ O/K ₂ O 1Al ₂ O ₃)	BAP (74B ₂ O ₃ 9Al ₂ O ₃ 17PbO)	BB (50B ₂ O ₃ 50Bi ₂ O ₃)	FP10 (10Sr(PO ₃) ₂ 35AlF ₃ 30CaF ₂ 15SrF ₂ 10MgF ₂)	FP20 (20Sr(PO ₃) ₂ 30AlF ₃ 22CaF ₂ 18SrF ₂ 10MgF ₂)
T_m (°C)	CVD	1500	1620	1300	1250	1000	1050
T_g (°C)	1120	560	530	480	420	440	480
α (10 ⁻⁷ K ⁻¹)	6	75	33	60	110	160	155
n_e	1.4602	1.5146	1.4733	1.6160	2.065	1.4602	1.5042
v_e	68	64	66	44	19	90	80
ρ (g/cm ³)	2.20	2.53	2.22	3.14	6.63	3.45	3.52
UV-edge (nm)	150	195	175	300	400	160	165
band gap (eV)	8.2	6.3	7.0	4.1	2.0	7.8	7.5
Δn^* (nm)	+50	-100	+60	100	45	-120	-190
$\sim \Delta n$ (10 ⁻⁴)	+10	-5	+6			-10	-50

Δn^* fs-laser induced difference of optical path length.

Δn fs-laser induced refractive index change measured with different methods.

correlation between the fs-laser-written changes and the glass composition and structure.

2. Experimental

Most of the used glasses were prepared with high purity raw materials by special techniques in the laboratory with high optical quality. Preparation and examination of the different glass samples were described in detail in [7–11]. Polished plates ($\sim 15 \times 20 \text{ mm}^2$, thickness 1–2 mm) of the different glasses were used for direct laser writing. A commercial Ti:sapphire femtosecond laser system (Spectra-Physics, Spitfire) which produces ~ 50 -fs pulses with a repetition rate of 1 kHz at a wavelength of 800 nm was used. To induce refractive index modifications pulses with varying energy, 0.1–10 μJ , were focused

100–200 μm below the surface of the polished glass plates using a microscope objective (10–20 $\times$, NA = 0.25–0.45). The sample was moved perpendicularly to the laser beam axis (Fig. 1) with a speed of 0.01–1 mm/s by a computer-controlled two-axis positioning system (Physik Instrumente, M126.DG). The fs-laser induced tracks were investigated with different methods: the refractive index profiles were directly measured with a refracted-near-field (RNF) profilometer (Rinck Elektronik, Germany), and indirectly on the difference of the optical path length (Δn in nm) by a special interference microscope with shearing technique (ZEISS, JENAPOL interphako with MICROFRINGE for JENAMAP). Additionally, light and electronic microscopic images were used.

3. Results and discussion

The selected glass compositions and some measured properties of the glass samples used for fs-laser writing experiments including the measured results of the induced refractive index changes are summarized in Table 1. Pure silica glass was used for comparison with known large positive Δn of the irradiated region [1–6]. The other glasses used are all interesting for special optical applications. They were prepared with ultra-high purity raw materials (without any refining agents) and so their measured UV absorption edge correlates with their intrinsic UV absorption or the optical band gap. Their structure and properties were studied intensively before [7–12]. BK7 and Duran are well known old borosilicate glass types. In UBK7 samples ($\approx$ BK7 with high UV-transmission) a negative Δn was induced, in UV-Duran a positive but lower value. The structure and properties

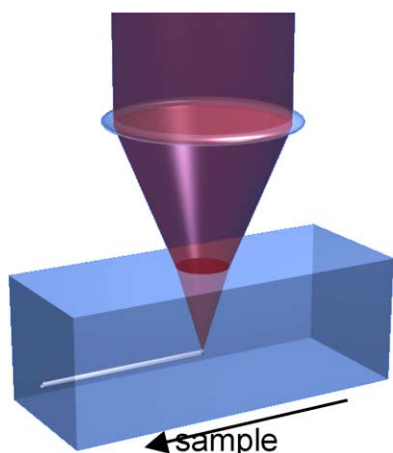


Fig. 1. Schematic diagram of the experimental setup.

of these borosilicate glasses are mainly determined by the M_2O/B_2O_3 and the B_2O_3/SiO_2 ratio. The high alkali

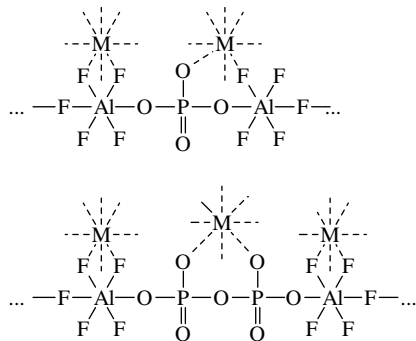


Fig. 2. Schematic models of FP glass structure with increasing phosphate content.

content of BK7 causes a high percentage of non-bridging oxygens and a lower UV band gap (~ 6.3 eV, $\lambda \sim 195$ nm). Structure units are tetrahedra of SiO_4 , BO_4 , few BO_3 triangles and many alkali ions. Duran has only a low alkali content. Structure units are tetrahedra of SiO_4 , few of BO_4 and AlO_4 , and many BO_3 triangles, connected with entirely bridging oxygens. This leads to an extreme intrinsic UV edge near 175 nm (~ 7.0 eV) and to a very low thermal expansion coefficient.

The composition of the BAP glass is similar to the optical glass KzFS1 whose refractive index is strongly dependent on the cooling rate, $\Delta_{10}n \sim 250 \times 10^{-5}$. Structure units are mainly BO_3 triangles, tetrahedra of AlO_4 and BO_4 , and PbO_x superstructural units. The large PbO content causes a small band gap (~ 4.1 eV, 300 nm) and we expected large induced refractive index changes. Although the obtained values are relatively

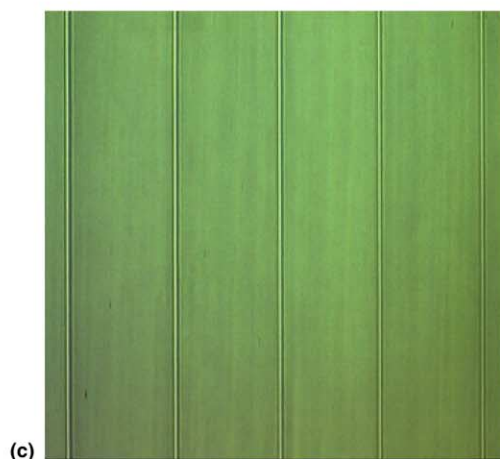
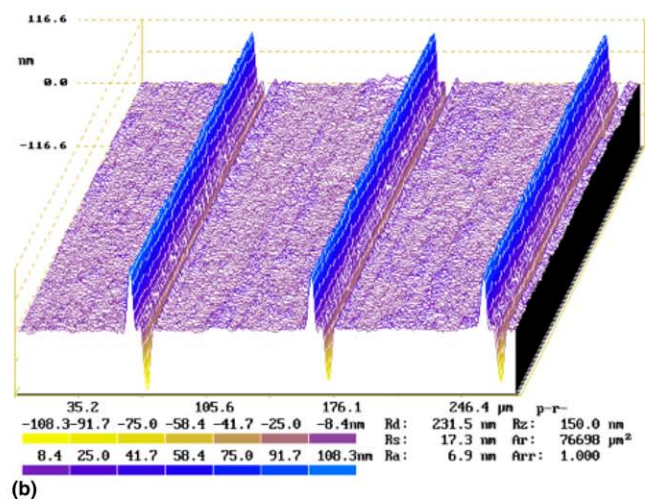
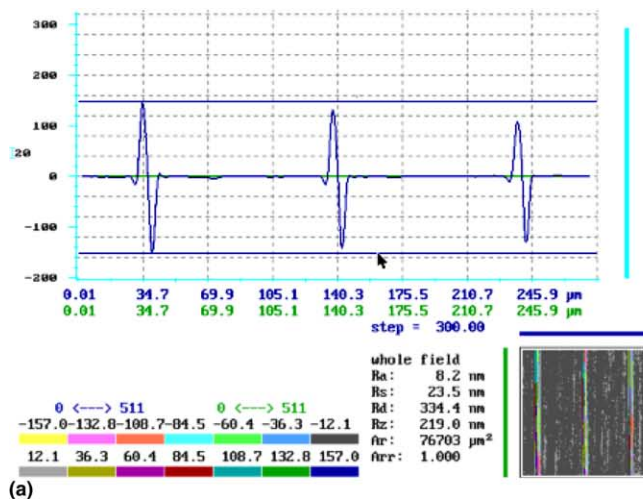


Fig. 3. Different images of the measured fs-written tracks in the volume of FP20 glass samples with interference microscope (Carl Zeiss JENAPOL.U.Interphako.map): (a) shear image (image and mirror image) of three induced tracks: the induced change of the optical path length increases from ~ 100 nm (right) to ~ 150 nm (left) with increasing pulse energy. That means, the induced refractive index modification, Δn , increases with increasing pulse energy. The distances between the induced tracks are $\sim 100 \mu m$. The optical microscope image of the three tracks is shown in the right corner on the bottom. (b) 3D shear image of refractive index modification of three induced tracks with an around 115 nm shows the induced line profiles. (c) Optical microscope image with differential interference contrast (DIC) of five induced tracks in $\sim 100 \mu m$ distances demonstrates the induced refractive index modification. The maximal measured value was $\Delta n \sim 5 \times 10^{-3}$.

low they are higher than in the BB glass sample which has the highest refractive index, $n \sim 2.1$, and the smallest band gap (~ 2.0 eV, 400 nm) due to the high Bi_2O_3 content. At present it is still questionable whether the fs-induced tracks in the two glasses have positive or negative Δn . Further investigations are necessary. The BB glass consists of BO_4 tetrahedra and BO_3 triangles and BiO_x superstructural units [12].

The last two glasses FP10 and FP20, in Table 1, which are totally different from the oxide glasses [7] showed surprisingly the largest fs-induced refractive index changes. The absolute values increased with increasing phosphate content and the change of the irradiated volume is unambiguously negative. The refractive index of these FP glasses is nearly independent on the cooling rate, $\Delta_{10}n \sim 40 \times 10^{-5}$, but strongly dependent on the phosphate content. Fig. 2 illustrates schematically the corresponding structural models. They consist of fluoroaluminate groups with embedded mono- and diphosphate units. Pure fluoroaluminate glasses, FP0, have a very low refractive index, $n_e \sim 1.40$, and a large band gap (~ 8.25 eV, 150 nm) which is comparable with those of pure SiO_2 and fluoride single crystals. But FP0 glasses have a very large tendency to crystallize due to the strong ionic fluorine bonds. Cooling rates, >100 K/min, are necessary for glass formation. Their optical properties are complementary to fluoride crystals, e.g. CaF_2 . The glass forming ability increases drastically with introducing small amounts of phosphates, due to the more covalent bonds of P–O preventing the crystallization of fluorides. The properties of FP glasses are mainly determined by the ratio of fluoride to phosphate groups. The band gap decreases, and the refractive index increases drastically, >1.50 , with increasing phosphate content. Fluoroaluminate glasses are much more stable against UV radiation than phosphate glasses. The generation of color center defects, which are mainly phosphorus–oxygen and impurity related, increases with increasing phosphate content. Experiments were carried out using UV-lamps, 248 nm–KrF excimer laser (ns- and fs-pulses) and 193 nm–ArF excimer laser (ns-pulses). Defect generation and healing behavior, using thermal treatment (temperature-dependence) and bleaching experiments (radiation-dependence), were investigated. The results of these experiments and the developed model enable the separation of induced color center absorption bands and the prediction of this defect generation depending on the used radiation source and glass composition. A two-step creation term including an energy transfer process and a one-photon bleaching term was assumed. The healing of strong color center defects at temperatures below T_g is postulated to be a diffusion-controlled process [7,8].

These laser-induced color center defects are only absorptive changes. It was not possible to detect any refractive changes under these previous conditions, in

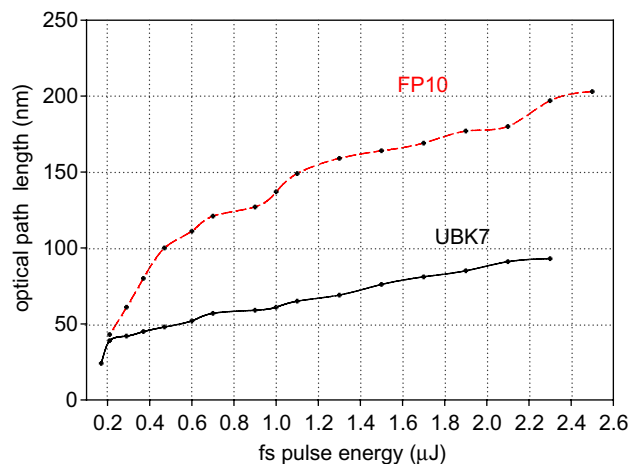


Fig. 4. Measured difference of the induced optical path length in the different glass samples FP10 and UBK7 depending on the pulse energy. Maximal values of (negative) Δn : FP10 $\sim 1 \times 10^{-3}$ and UBK7 $\sim 5 \times 10^{-4}$.

contrast to this work where sharp refractive changes were detected with different methods. These are illustrated in Figs. 3–5. Very sharp tracks are fs-written inside the volume of the sample plates (Fig. 3). The induced changes are depending on glass composition, pulse energy and writing speed (Figs. 4 and 5). The largest refractive index changes were measured with different methods in the FP20 glass sample with $\Delta n \sim 5 \times 10^{-3}$.

Formation of micro-cracks was found with increasing pulse energy, $>6 \mu\text{J}$ (Fig. 5(b)). The laser-induced tracks showed different chemical etching behavior. In a FP10 glass sample the etching rate with an aqueous solution of a HF/HNO_3 mixture from the track was much higher than its environment (Fig. 5(c)).

4. Conclusions

It could be shown with these experiments that the difference of the induced refractive index is strongly dependent not only on the laser writing conditions but also on the glass composition and its microstructure. Two well-known borosilicate glasses, BK7 and Duran, have shown contrasting induced Δn behavior due to their contrasting microstructure. The negative value in BK7 could be obtained by movement of the alkali ions accompanied with a redistribution of non-bridging and bridging oxygens. The positive value of Duran (with low alkali content) could be caused by a redistribution of the connectivity of BO_3 triangles and SiO_4 tetrahedra, which was also found in this glass type depending on the cooling rate [9] or by phase separation.

The large induced Δn values in FP glasses depend strongly on the micro-glass-structure and its thermodynamic and kinetic behavior. It could be correlated with a redistribution of the connectivity between fluoride and

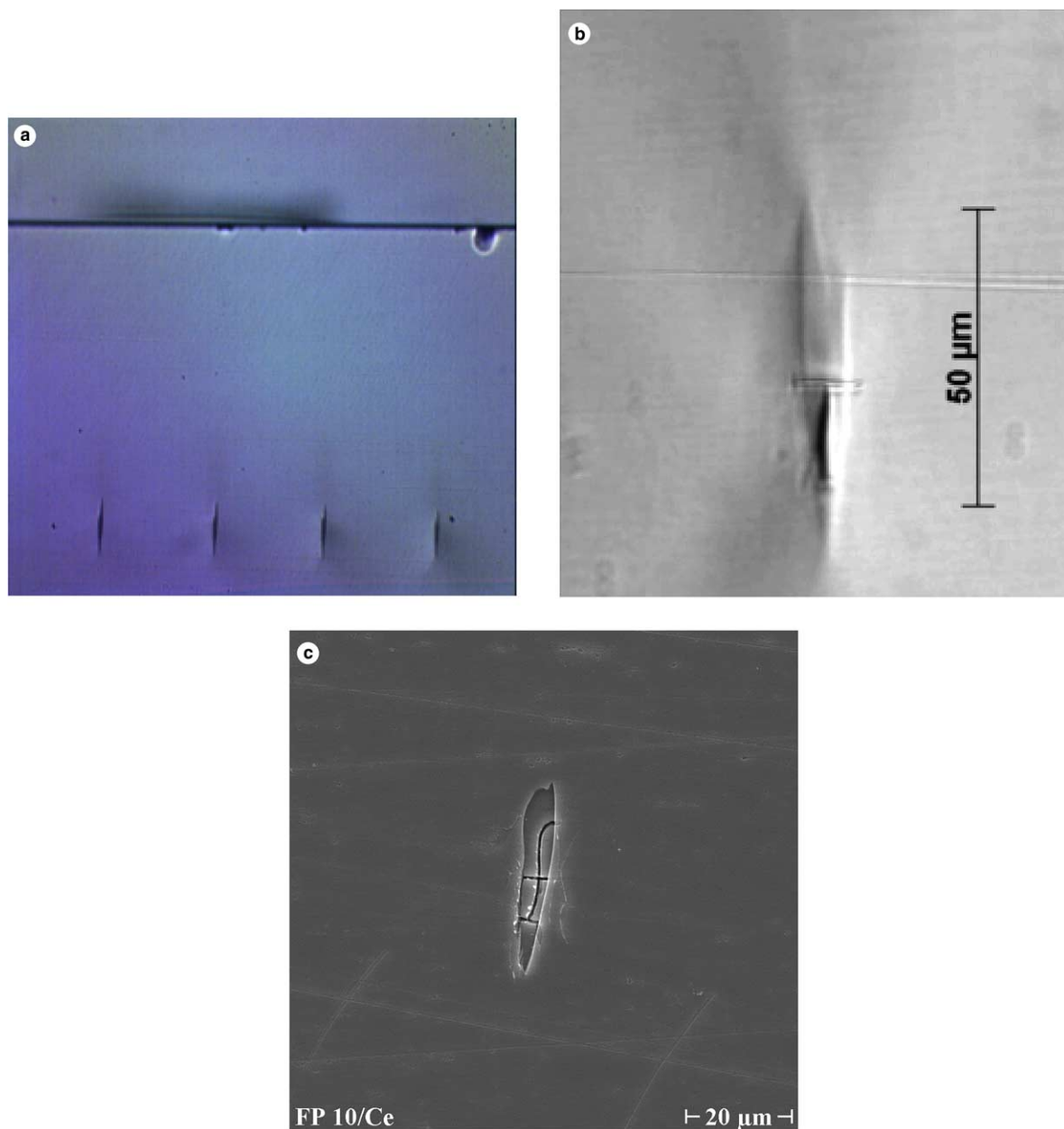


Fig. 5. Images of cross sections of the fs-laser written tracks in the volume of FP glass samples demonstrated with different methods: (a) with a light microscope (objective 10×), cross sections of four tracks induced 200 μm below the surface of the glass plate; distances between the tracks ~100 μm; height of the tracks 20–30 μm; width ~4–7 μm; increase with increasing pulse intensity; measured maximal value $\Delta n \sim 5 \times 10^{-3}$. (b) With electron microscope: micro crack formation with 80-fs pulses, ~6 μJ, in FP20 sample. (c) SEM micrograph after chemical etching of a FP10 sample: the etching rate of the fs-irradiated volume is much higher than the etching rate of its environment.

phosphate groups in a similar way found earlier for the crystallization behavior of these glasses: the connection between fluoride and phosphate groups is broken again leading to formation of fluoride and phosphate crystals [7]. The fs-irradiated volume should be more rich of fluoride groups which causes a lower refractive index.

The glassy state is from the point of classical thermodynamics an unstable state which was frozen in on the way to the equilibrium (crystals). The fs-laser treatment of a localized glass volume can shift this state. So, the

fs-laser writing technique in combination with the large variety of glass composition and local structure is an excellent new tool to fabricate new integrated optical devices.

References

- [1] K. Miura, J. Qiu, H. Inouye, T. Mitsuyu, K. Hirao, Appl. Phys. Lett. 71 (1997) 3329.

- [2] K. Hirao, K. Miura, J. Non-Cryst. Solids 239 (1998) 91.
- [3] J.W. Chan, T.R. Huser, S.H. Risbud, D.M. Krol, Appl. Phys. A (2002).
- [4] A.M. Streltsov, N.F. Borelli, J. Opt. Soc. Am. B 19 (2002) 2496.
- [5] A.M. Streltsov, N.F. Borelli, Opt. Lett. 26 (2001) 42.
- [6] M. Will, S. Nolte, B.N. Chichkov, A. Tünnermann, Appl. Opt. 41 (2002) 4360.
- [7] D. Ehrt, P. Ebeling, U. Natura, J. Non-Cryst. Solids 263 & 264 (2000) 240.
- [8] U. Natura, D. Ehrt, J. Nucl. Instr. Meth. B 174 (2001) 143, and 151.
- [9] D. Möncke, D. Ehrt, H. Eckert, V. Mertens, Phys. Chem. Glasses 44 (2003) 113.
- [10] D. Ehrt, P. Ebeling, Glass Technol. 44 (2003) 46.
- [11] D. Ehrt, Glass Technol. 41 (2000) 182.
- [12] C.E. Stone, A.C. Wright, R.N. Sinclair, S.A. Feller, M. Affatigato, D.L. Hogan, N.D. Nelson, C. Vira, Y.B. Dimitriev, E.M. Gattef, D. Ehrt, Phys. Chem. Glasses 41 (2000) 409.